Engineered Antimicrobial Peptide with Collagen Self Assembling Property for Restoring Oral Health

Antranig Baghdassarian[1],Nur Ceren Kabakci[2], Aya Cloyd[2], Kyle Boone[2,3], Paulette Spencer[2,3,4], and Candan Tamerler[2,3,4]

[1]Meinig School of Biomedical Engineering, Cornell University; [2]Bioengineering Program, University of Kansas; [3]Institute for Bioengineering Research, University of Kansas; [4]Department of Mechanical Engineering, University of Kansas

Current Dental Pandemic

Oral diseases, such as dental caries and periodontal disease are costly to treat and affect millions around the world[1,2].The role of oral bacteria is recognized as one of the leading causes for oral diseases. *S. Mutans as one of the major player in the oral microbiome play a critical role in initiating the diseases leading to d*ental restoration failures as they attack the adhesive/dentin interface (Figure 1). As the biofilm develops, the restoration as well as the remaining tooth structure gets jeopardized via demineralization [2,3,4].

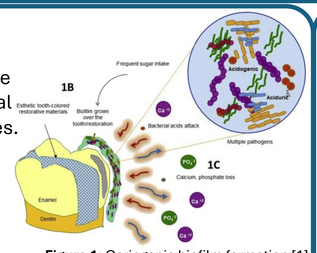


Figure 1: Cariogenic biofilm formation [1]

Current restoration face limitations due to relatively short life span, resulting in secondary caries. Due to demineralization, collagen fibrils are more vulnerable. Antimicrobial peptides are promising agents to address antibiotics resistance [2,3,4]. Our research focuses on testing the efficacy of a bifunctional peptide to address both both issues at this vulnerable interface by combining collagen binding and antimicrobial domains[3,5], separated by a spacer: CBP-sp-AMP. This bifunctional design aims to provide localized antimicrobial protection within the collagen of the demineralized dentin preventing recolonization and thus improve the longevity of dental restorations.

Methodology

Peptide Synthesis

Solid Phase Peptide Synthesizer with Rink resin using Fmoc chemistry via an automated peptide synthesizer.





Atomic Force Microscopy (AFM) Imaging

Bruker AFM 8 Multimode on Tapping mode used to depict the surface of collagen and monitor the peptides' affect on collagen fibril assembly.

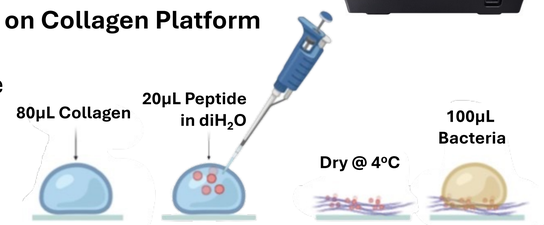
Antimicrobial Activity Assay in Solution

- Serial dilution of peptide in DI H_2O for a range of concentrations incubated with S. Mutans.
- Viability was measured via AlamarBlue assay using a Cytation™ 3.
- Calculate minimum inhibitory concentration (MIC)



Antimicrobial Activity Assays on Collagen Platform

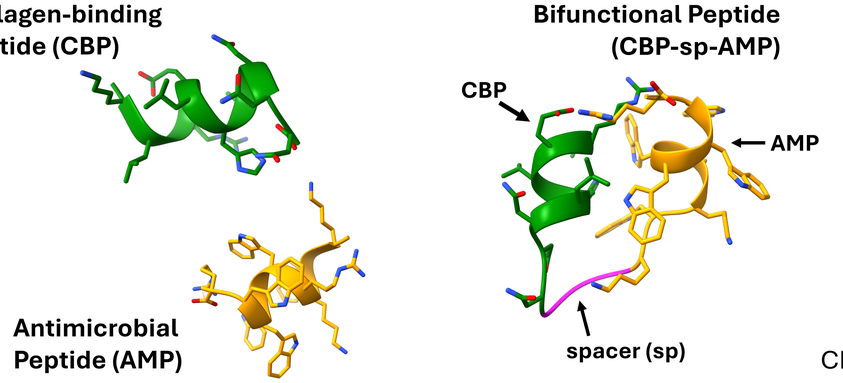
Assess antimicrobial assay efficacy of bifunctional peptide when assembled within type I collagen platform.



Results

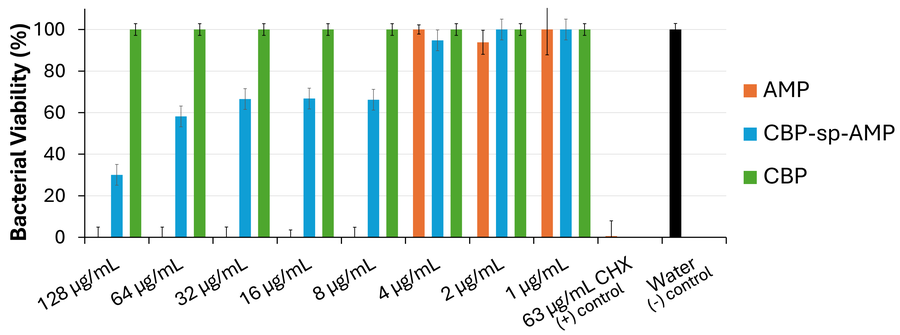
Peptides' Structure

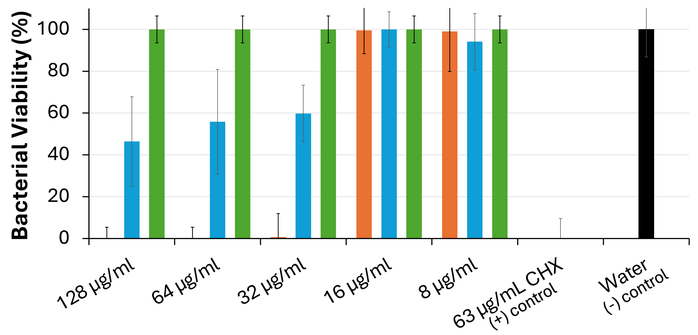
Sequence structures generated in PEPFOLD3 & mapped using UCSF ChimeraX.



Bifunctional Peptide Antimicrobial Activity:

CBP-sp-AMP shows overall diminished antimicrobial properties with a significant MIC shift compared to AMP alone.

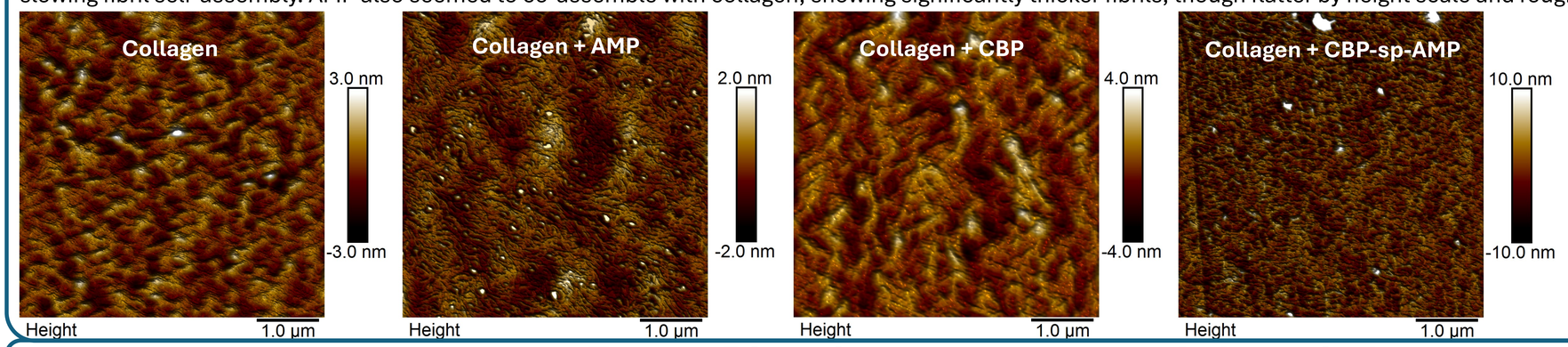




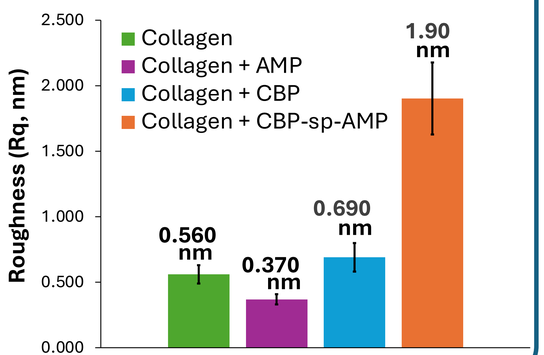
CBP-sp-AMP lowers the bacteria viability to ~50% in-solution with 8ug/mL to 64ug/mL dosage. In the collagen platform it lowers bacteria viability to ~50% at with 32ug/mL to 64ug/mL dosage.

Affect of Peptide on Collagen Fibril Assembly:

The incorporation of peptides directly affect the collagen fibril assembly. CBP increased fibril widths compared to collagen whereas CBP-sp-AMP showed smaller fibril widths compared to collagen, potentially slowing fibril self-assembly. AMP also seemed to co-assemble with collagen, showing significantly thicker fibrils, though flatter by height scale and roughness.



Average roughness values over co-assembled peptide-collagen surfaces seem to correspond to density of fibril formation respectively.



Conclusions

The activity of the respective domains within the bifunctional peptide were affected when joined by the spacer.

- CBP-sp-AMP showed a decrease in its antimicrobial activity, though its activity remained consistent when assessed co-assembled with collagen.
- When co-assembled with collagen, CBP-sp-AMP appeared to slow the collagen fibril assembly, leading to a network of thinner fibrils formed, though a denser network than with respective domains individually.

Future Directions

- **Using machine learning methods**, improving spacer design & testing different peptide combinations to improve respective domain functionality.
- Examining the **nanomechanical changes in fibril assembly**, potentially over time-lapse.
- Observe affects on fibril orientation/alignment due to peptide influence on collagen fibril assembly.



Acknowledgments

- The National Science Foundation Award No. 2243816
- Department of Chemical & Petroleum Engineering – University of Kansas
- Research reported in this poster was supported by the National Institute of Dental & Craniofacial Research of the National Institutes of Health under Award Number R01DE025476.
- Thank you to the Tamerler Research Group for all your help.

References

[1] Balhaddad, Abdulrahman A., *et al.* "Toward dental caries: Exploring nanoparticle-based platforms and calcium phosphate compounds for dental restorative materials." *Bioactive materials* 4 (2019): 43-55.
[2] Sarikaya, Rizacan, et al. "Bioinspired multifunctional adhesive system for next generation bio-additively designed dental restorations." *Journal of the mechanical behavior of biomedical materials* 113 (2021): 104135.
[3] Xie, S.-X.; *et al.* Antimicrobial Peptide–Polymer Conjugates for Dentistry. *ACS Applied Polymer Materials 2020*, *2* (3), 1134–1144.
[4] Cloyd, Aya K., et al. "Engineered peptides enable biomimetic route for collagen intrafibrillar mineralization." *International Journal of Molecular Sciences* 24.7 (2023): 6355.
[5] Yuca, Esra, et al. "Reconfigurable dual peptide tethered polymer system offers a synergistic solution for next generation dental adhesives." *International Journal of Molecular Sciences* 22.12 (2021): 6552.